MILITARY SPECIFICATION

BODY ASSEMBLY: 10516576
(PERISCOPE: M32C AND M32CE1)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a modular unit of the Tank Gunner's Periscopes M32C and M32CE1, herein referred to as "body assembly".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS (See 6.4)

US Army Armament Research and Development Command (ARRADCOM)

8565556	Pressure Tester, Two-Station
10516576	Body Assembly
11732109	Gage Interchangeability, Maximum
11732110	Gage Interchangeability, Minimum

PACKAGING DATA SHEET

10516576	Packaging of Body Assembly 10516576

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. The body assembly shall be manufactured in accordance with Drawing 10516576 and drawings pertaining thereto.

3.2 Interchangeability (body assembly to head assembly). The body assembly shall be capable of being inserted into, then removed from its head assembly without restriction (gages are specified in lieu of head assembly module in 4.6.10).

3.3 General specification. The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926:

a. Dimension and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking

3.4 Environmental.

3.4.1 Storage temperature. The body assembly shall show no evidence of cement separation, glass breakage, bonding failure, or other physical failure, and shall meet all the requirements of this specification at standard ambient temperature (73^{0} + 18^{0}F) after having been exposed and thermally stabilized at ambient temperatures of -80^{0} to + 160^{0}F.

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3.4.2 Operating temperatures. The body assembly shall meet the applicable requirements of 3.5 while exposed and thermally stabilized at ambient temperatures of -40° and +150°F.

3.5 Operability.

3.5.1 Disengagement of boresight knob locks. The direct force required to disengage each boresight knob from the lock position shall be within the limits of 2.0 to 5.0 pounds (lbs) when checked at standard ambient temperature (73° ± 18°F). When checked at the temperature conditions specified in 3.4.2, the direct force shall not exceed 9.0 lbs.

3.5.2 Rotation of boresight knobs. The running torque required to rotate each boresight knob through the range specified in 3.8.3 shall be within the limits of 0.5 and 5.0 pound inches (lb-ins) when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 8.0 lb-ins.

3.5.3 Diopter movement. The running torque required to rotate the diopter movement through the range specified in 3.14.2 shall be within the limits of 1.5 and 6.0 lb-ins when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 15 lb-ins.

3.5.4 Focusing lock. This requirement shall be met with a 10 pound inches (lb-ins) torque applied to the locking screw on Housing 11727432. There shall be no rotation of Eccentric 8205635 when a 48 oz-in torque is applied to the eccentric.

3.5.5 Objective focusing mechanism. This requirement shall be met with the locking screw backed off a minimum of two full revolutions from the locked position established in 3.5.4. The running torque required to rotate Eccentric 8205635 through a minimum of one full revolution in each direction shall not exceed 30 oz-ins when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 50 oz-ins.

3.6 Orientation. The requirements of 3.7, 3.8 and 3.10 through 3.16 inclusive shall be met with the body assembly eyepiece focused on the instrument reticle.

NOTE: The following references apply throughout this specification.
Reference Line "Y", see Drawing 10516576.
Datum Plane "X" (Intersection of planes).

3.7 Parallax. Parallax between the reticle and the image of a target at 1200 meters shall not exceed 0.1 mil as adjusted by Eccentric 8205635.

3.8 Collimation. The following requirements shall be met prior to meeting the vibration and shock requirements of 3.9.

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3.8.1 Pre-vibration/pre-shock setting. An incident line of sight shall be provided which is perpendicular to Datum Plane "X" in deflection within 0.20 mil, and which is elevated 20.0 mils ± 0.20 mil "above" the deflection plane as defined. The deflection direction is defined by the plane which is perpendicular to both Datum Plane "X" and Reference Line "Y". The term "above" indicates that the incoming line of sight is rotated towards the instrument eyepiece.

3.8.2 Reticle movement. Reticle (boresight cross) movement shall be measured against an external target from the collimated setting of 3.8.1 to meet the requirements of 3.8.2.1 and 3.8.2.2.

3.8.2.1 Elevation. Clockwise rotation of the elevation knob shall move the reticle, in a direction towards the knob, a minimum of 8.5 mils from the collimated setting. Counterclockwise rotation of the elevation knob shall move the reticle, in a direction away from the knob, a minimum of 5.5 mils from the collimated setting.

3.8.2.2 Deflection. Clockwise rotation of the deflection knob shall move the reticle, in a direction towards the knob, a minimum of 7.0 mils from the collimated setting. Counterclockwise rotation of the deflection knob shall move the reticle, in a direction away from the knob, a minimum of 7.0 mils from the collimated setting.

3.8.3 Knob rotation. Continuous movement of reticle shall be evident until limited by positive engagement of clockwise and counterclockwise stops in each boresight knob.

3.8.4 Line of sight variation. Variation in the line of sight in elevation, as the deflection knob is rotated by at least 7.0 mils in each direction from the position established in 3.8.1, shall not exceed 0.1 mil (total variation not exceed 0.2 mil).

3.8.4.1 Reticle shift. Disengagement or reversal of either knob shall not cause the line of sight to shift by more than 0.1 mil in a direction perpendicular to the intended motion.

3.8.5 Boresight backlash. Backlash in each boresight mechanism shall not exceed 0.1 mil.

3.9 Vibration/Shock.

3.9.1 Vibration "A". Vibration shall be applied along each of three mutually perpendicular axes of the body assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be .5G ± 10%.

b. During a frequncy sweep of 5 to 9 Hz, increase acceleration linearly from .5 G to 1.5 G's ± 10%.

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c. During a frequency sweep of 9 to 500 Hz, maintain constant acceleration of 1.5 G's ± 10%.

d. During a frequency sweep of 500 to 9 Hz, maintain constant acceleration of 1.5 G's ± 10%.

e. During a frequency sweep of 9 to 5 Hz, decrease acceleration linearly from 1.5 G's to .65 G ± 10%.

f. At 5 Hz, the acceleration shall be .5 G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis. The body assembly shall show no evidence of physical failure subsequent to vibration. Following this vibration test the body assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.18 inclusive.

3.9.2 Vibration "B". Vibration shall be applied along each of three mutually perpendicular axes of the body assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be 1.0 G ± 10%.

b. During a frequency sweep of 5 to 500 Hz, maintain constant acceleration of 1.0 G ± 10%.

c. During a frequency sweep of 500 to 5 Hz, maintain constant acceleration of 1.0 G ± 10%.

d. At 5 Hz, the acceleration shall be 1.0 G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of 15 minutes ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes ± 3 minutes). The body assembly shall show no evidence of physical failure subsequent to vibration. Following this vibration test the body assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.18 inclusive.

3.9.3 Shock. The body assembly shall be subjected to 24 cumulative shocks consisting of three each at 35 G's ± 10%, then 20 G's ± 10% applied in each direction along each of the two mutually perpendicular horizontal axes (all four sides). The shock pulse shall have a half sine wave with a time duration of 3 ± .2 milliseconds. The body assembly shall also be subjected to six cumulative shocks consisting of three each at 35 G's ± 10%, then 20 G's ± 10% applied in one direction along the vertical axis (normal

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operating position of the body assembly). The shock pulse shall have a half sine wave with a time duration of 18 + 3 milliseconds. The body assembly shall show no evidence of physical failure subsequent to shock. Following the shock test, the body assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.18 inclusive.

3.10 Collimation change due to vibration/shock. This requirement shall be met subsequent to meeting the requirement of 3.9. The incident line of sight previously established shall coincide within 0.1 mil (including test equipment error) with the instrument reticle.

3.11 Image tilt. The image of a target line which is originally perpendicular to Reference Line "Y" shall emerge perpendicular to Reference Line "Y" within 1.0 degree of arc.

3.12 Parallelism of reticle and image. The image of a target line which is originally perpendicular to Reference Line "Y" shall emerge parallel within 30 minutes of arc to the horizontal reticle line.

3.13 Objective focusing range. Rotation of Eccentric 8205635 shall cause the objective assembly to travel a minimum of 0.020 inch in each direction from the position established in 3.7.

3.14 Dioptral adjustment.

3.14.1 Zero setting. This requirement shall be met while observing the projected image through a dioptometer set for zero diopter. The diopter scale shall indicate zero within 1/4 diopter when the reticle is brought into focus.

3.14.2 Excursion range. The diopter movement shall be adjustable through a minimum range of -3.0 to +3.0 diopters as read on the diopter scale.

3.14.3 Diopter backlash. Backlash in the diopter movement shall not exceed 1/8 diopter as read on the diopter scale.

3.15 Resolution. The resolution shall be 6.0 seconds of arc or less when viewing with an auxiliary telescope having magnification of at least 3.0 power. Resolution consists of detectable line structure and proper line count in all four meridians. Eyepiece focus shift between any two meridians shall not exceed 0.25 diopter.

3.16 Reticle illumination. With a 24 volt direct current source energizing the reticle light, the markings shall be visable when observed in a dark room.

3.17 Sealing. The body assembly shall show no evidence of leakage in excess of 0.2 pounds per square inch gage (psig) when subjected to an internal pressure of 5 + 0.10 psig for a minimum of 2 hours. The interior atmosphere of the accepted body assembly shall contain nitrogen having a dew point at least as low as minus 25°F at atmospheric pressure. The body assembly shall then be sealed with the dry nitrogen specified to an internal pressure of between 0.25 and 0.50 psig.

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3.18 Cleanliness.

a. Defects. There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations on any glass component when viewing through either end.

b. Dirt. Inspection for dirt and foreign particles (classed as dirt) shall be made from the eyepiece end only. The maximum size of any particle appearing on the reticle shall be a reticle line width. The maximum number of dirt particles appearing within a central field of view of 50 mils diameter shall be three. The maximum number of dirt particles appearing outside this field of view shall be three. The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution.

3.19 Workmanship. The body assembly shall meet the workmanship requirements of MIL-F-13926 and the applicable specifications listed in the Technical Data Package (TDP). The optical surfaces of the body assembly shall be free from scratches or blemishes that may affect their optical qualities, or appearance. In addition, the exterior protective surface finishes of the body assembly shall be free of defects that may affect their protective qualities or appearance.

3.20 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inpsections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Classification of inspections. Unless otherwise specified herein or in the contract, the following types of inspections shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. Unless otherwise specified, the sample for first article approval tests shall consist of three body assemblies plus three each of all items covered by Supplementary Quality Assurance Provisions (SQAP). The responsibility (Government or Contractor) for first article testing shall be as specified in the contract.

4.3.2 Inspections to be performed.

4.3.2.1 Body assemblies. First article assemblies shall be tested as specified herein for all the requirements of this specification, except 3.9.2.

4.3.2.2 SQAP items. The items covered by SQAP shall be tested as specified thereon for all characteristics delineated on the SQAP.

4.3.3 Rejection. If any body assembly or SQAP item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any component, subassembly or assembly in th sample to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.4.2 Examination. The examinations listed herein for classification of defects shall be performed on inspection lots as defined in 4.4.1. of this specification.

4.4.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the Supplementary Quality Assurance Provisions (SQAP) listed in the Technical Data Package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4.2.2 Body assemblies. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, Inspection Level II, and the sampling plans specified in Tables I, II, III and IV herein. Examination for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 100 percent inspection. The requirements and tests in Table I shall be inspected on a 100% basis.

TABLE I - CLASSIFICATION OF DEFECTS
(100 Percent Inspection)

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:			
101.	Interchangeability, body assembly to head assembly	3.2	4.6.10
102.	Line of sight variation	3.8.4	4.6.5.4
103.	Reticle shift	3.8.4.1	4.6.5.5
104.	Dioptral adjustment, zero setting	3.14.1	4.6.5.10.1
105.	Resolution	3.15	4.6.6
106.	Reticle illumination	3.16	4.6.7
MINOR:	NONE		

NOTE: The tests in Table I shall be conducted at a temperature of $73^{o} \pm 18^{o}F$.

TABLE II - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% DEFECTIVE		
107.	Operability, disengagement of boresight knob locks	3.5.1	4.6.3.1
108.	Operability, rotation of boresight knobs	3.5.2	4.6.3.2
109.	Operability, diopter movement	3.5.3	4.6.3.3
110.	Operability, focusing lock	3.5.4	4.6.3.4
111.	Operability, objective focusing mechanism	3.5.5	4.6.3.5
112.	Collimation, pre-vibration/ pre-shock setting	3.8.1	4.6.4.1
113.	Vibration "B"	3.9.2	4.6.4.3
114.	Collimation change due to vibration	3.10	4.6.4.6
115.	Parallax	3.7	4.6.5.1

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TABLE II - CLASSIFICATION OF DEFECTS (Continued)

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 0.65% DEFECTIVE		
116. Reticle movement, elevation	3.8.2.1	4.6.5.2
117. Reticle movement, deflection	3.8.2.2	4.6.5.2.1
118. Knob rotation	3.8.3	4.6.5.3
119. Boresight backlash	3.8.5	4.6.5.6
120. Image tilt	3.11	4.6.5.7
121. Parallelism of reticle and image	3.12	4.6.5.8
122. Objective focusing range	3.13	4.6.5.9
123. Dioptral adjustment, excursion range	3.14.2	4.6.5.10.2
124. Diopter adjustment, backlash	3.14.3	4.6.5.11
125. Sealing	3.17	4.6.8
126. Cleanliness	3.18	4.6.9
127. Workmanship	3.19	Visual-Tactile
MINOR: NONE		

NOTE: The tests in Table II shall be conducted at a temperature of $73^{o} \pm 18^{o}F$.

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Special sampling.

4.5.1 General. One body assembly shall be selected at random as a special sampling from each 100 produced or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.2	Applicable Drawings, Visual
302.	General specification	3.3	MIL-F-13926, MIL-O-13830, Visual

NOTE: The tests in Table III shall be conducted at a Temperature of $73^{o} \pm 18^{o}F$.

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4.5.2 Environmental. Three body assemblies shall be selected at random as samples from each 50 produced, or from each months production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall then meet the requirements and tests in Table I.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures (-80° to +160°F)	3.4.1	4.6.1
304.	Operability, disengagement of boresight knob locks	3.4.2, 3.5.1	4.6.2, 4.6.3.1
305.	Operability, rotation of boresight knobs	3.4.2, 3.5.2	4.6.2, 4.6.3.2
306.	Operability, diopter movement	3.4.2, 3.5.3	4.6.2, 4.6.3.3
307.	Operability, focusing lock	3.4.2, 3.5.4	4.6.2, 4.6.3.4
308.	Operability, objective focusing mechanism	3.4.2, 3.5.5	4.6.2, 4.6.3.5

NOTE: The tests for Characteristics 304 through 308 inclusive, shall be conducted at -40° and at +150°F.

4.5.3 Verification follow-up. Three body assemblies shall be randomly selected as samples from production approximately at the 25 percent, 50 percent and 75 percent completion interval of each production order or contract. Each sample shall meet the requirements and tests in Table V.

TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
309.	Collimation, pre-vibration setting,	4.8.1	4.6.4.1
310.	Vibration "A"	3.9.1	4.6.4.2
311.	Collimation change due to vibration	3.10	4.6.4.6
312.	Collimation, pre-shock setting	3.8.1	4.6.4.1
313.	Shock	3.9.3	4.6.4.4
314.	Collimation change due to shock	3.10	4.6.4.6

NOTE: The tests in Table V shall be conducted at a temperature of 73° ± 18°F.

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4.5.4 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved (see 4.4.3).

4.5.5 Inspection equipment. Requirements for inspection equipment to perform examinations and tests are specified herein and in specific contract clauses as applicable. See Section 6 of this specification for general provisions applicable to inspection designs.

4.5.6 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.7 Contractor furnished inspection equipment.

4.5.7.1 Testing equipment. In conjunction with 4.5.6, the special inspection equipment in 4.5.7.1.1 and 4.5.7.1.2 shall be utilized in performing the applicable tests of 4.6.

4.5.7.1.1 Sealing. The pressure testing equipment to test sealing of the body assembly shall conform to the internal pressure test requirements of Drawing 8565556, Pressure Tester-Two Station.

4.5.7.1.2 Performance. Special test equipment to test performance of the body assembly shall conform to the following requirements:

a. The testing equipment shall be capable of securely mounting the body with Datum Plane "X" in a vertical plane and Reference Line "Y" in a horizontal plane within 10 seconds. The necessary mounting features (keyways and locating surfaces) shall be incorporated into the testing equipment.

b. A target projector collimator having a clear aperture of at least 2.0 inches shall be a part of the test equipment. The target projector collimator shall be positioned and mounted so that the optical axis of the collimator is in alinement with the line of sight emitting from the central aperture (geometric center) of the body assembly objective. The reticle of target projector collimator shall have an appropriate calibrated crossline target test pattern containing a vertical line at least 14 mils in length (7 mils on each side of the horizontal reticle line) and 20 mils right of the geometric axis to determine compliance with the optical requirements herein. The vertical reticle line of the target test pattern shall be adjusted plumb. The optical axis of the target projector collimator shall be alined parallel with respect to Reference Line "Y" and autocollimated within 10 seconds to Datum Plane "X" of the body assembly.

c. A calibrated observation telescope of at least three power (3X) shall be a part of the test equipment. The observation telescope shall be positioned and mounted so that the optical axis of the telescope is in alinement with the line of sight emitting from the central aperture (geometric center) of the body assembly eyepiece. The reticle of the telescope shall

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have a crossline reticle pattern with tolerance lines to determine image tilt. The vertical reticle line of the telescope shall be adjusted plumb. The optical axis of the telescope shall be alined to the target projector collimator and autocollimated to Datum Plane "X" within 10 seconds.

d. Autocollimation and alinement shall be accomplished with the body assembly removed from the line of sight through the target projector collimator and the observation telescope. A set-up device containing a penta prism having a deviation of ± 5 seconds (calibrated) shall be utilized for alinement and autocollimation of the target projector collimator and the observation telescope. The necessary mounting features (keys and locating surfaces) shall be incorporated in the set-up device.

e. All set-up and calibrated equipment with complete provisions necessary for certification and proof of accuracy of the test equipment shall be provided and become a part of the testing equipment.

f. Upon completion of the fixture set-up and alinement procedures, the observation telescope shall be capable of being removed from the optical line of sight of the test equipment when not required in the performance tests.

4.6 Methods of inspection.

4.6.1 Storage temperature. This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope, this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Conditions" specified in the "General Requirement" of MIL-STD-810 and the conditions of 3.4.1. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see note below) to -80°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see note below) to +160°F. The body assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (73° ± 18°F), and the body assembly removed from the test chamber. The body assembly shall then be subjected to the visual, tactile and audible examination of 4.6. The body assembly shall conform to the requirements of 3.4.1.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperatures. This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope, this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Conditions" specified in the "General Requirements" of MIL-STD-810 and the conditions of

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3.4.2. The body assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see Note in 4.6.1) to -40°F. The body asembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at -40°F, the body assembly shall be subjected to the tests in 4.6.3.1 to 4.6.3.5, inclusive to determine compliance with the applicable requirements of 3.5.1 to 3.5.5 inclusive, for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see Note in 4.6.1) to +150°F. The body assembly shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period and while at +150°F, the body assembly shall again he subjected to the test in 4.6.3.1 to 4.6.3.5 inclusive, to determine compliance with the applicable requirements of 3.5.1 to 3.5.5 inclusive, for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see Note in 4.6.1) to room ambient temperature (73° ± 18°F) and the body assembly removed from the test chamber. Body assemblies which have passed the above tests shall be subjected to the test in 4.6.4.

4.6.3 Operability.

4.6.3.1 Disengagement of boresight knob locks. Standard measuring equipment capable of measuring the force values specified in 3.5.1 shall be utilized for this test. The force required to disengage each boresight knob from the lock position shall conform to the applicable requirements of 3.5.1 for the temperatures indicated.

4.6.3.2 Rotation of boresight knobs. Standard torque measuring equipment capable of measuring the torque values specified in 3.5.2 shall be utilized for this test. An appropriate adapter, to accommodate the boresight knobs shall be used with the torque measuring equipment. Torque shall be measured while rotating each boresight knob through the range specified in 3.14.2. The running torque required to rotate each boresight knob shall conform to the applicable requirements of 3.5.2 for the temperatures indicated.

4.6.3.3 Diopter movement. Standard torque measuring equipment capable of measuring the torque values specified in 3.5.3 shall be utilized for this test. An appropriate adapter, to accommodate the boresight knobs shall be used with the torque measuring equipment. Torque shall be measured while rotating the diopter movement through the range specified in 3.14.2. The running torque required to rotate the diopter knob shall conform to the applicable requirements of 3.5.3 for the temperatures indicated.

4.6.3.4 Focusing lock. This test shall be performed subsequent to the test in 4.6.5.9. A dial indicator, graduated in increments of .0001 of an inch, mounted on an appropriate stand, the supporting medium specified in 4.6.5.9 and a torque wrench capable of applying the specified torque values in 3.5.4 and 3.5.5, together with a standard type driver bit that properly fits the screw slot of the locking screw and eccentric, shall be utilized for performing this test. Secure the body assembly firmly on the supporting medium. With the specified torque applied to the locking screw, position the dial indicator so that its contact point touches the face of the movable objective tube of the daylight body and set the dial to read zero.

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Apply, in a clockwise direction, the specified torque to the eccentric, then observe the dial indicator reading. Then apply, in a counterclockwise direction, the specified torque to the eccentric and again observe the dial indicator reading. To conform to the requirements of 3.5.4, there shall be no change in the established dial indicator zero reading when observed after each torque application.

4.6.3.5 Objective focusing mechanism. Perform this test utilizing the test equipment specified in 4.6.3.4 except the dial indicator is not required. The locking screw shall be backed off a minimum of two full revolutions from its locked position. Torque shall be measured while rotating the eccentric through the range specified in 3.5.5. The running torque required to rotate the eccentric shall conform to the applicable requirements of 3.5.5 for the temperatures indicated. Upon completion of this test, the parallax adjustment of 4.6.5.1.1 shall be repeated.

4.6.4 Vibration/Shock.

4.6.4.1 Collimation, pre-vibration/pre-shock setting. This test and the following test of 4.6.4.6 shall be performed with the body assembly positioned and adjusted as in 4.6.5. While viewing through the body assembly eyepiece, position the body assembly reticle boresight mark (intersecting point of the horizontal and vertical reticle lines) coincident with the intersecting point of the vertical target line and the horizontal target line, located 20 mils from the geometric axis of the target reticle, by means of the boresight knobs. When coincidence has been established, slip the scales of the boresight knobs to a setting of (4) and recheck to insure coincidence is not disturbed. Record these settings for use in the test in 4.6.4.6. Without disturbing the boresight knob scale settings of (4), the body assembly shall then be subjected to the test in 4.6.4.2.

NOTE: Insure that knob scales are not disturbed (re-slipped) from their original setting of (4) throughout remainder of tests herein.

4.6.4.2 Vibration "A". This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope, this vibration test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the body assembly shall be subjected to the test procedure specified in 4.6.4.1 to determine compliance with 3.8.1. Subsequent to the test of 4.6.4.1 the body assembly shall be vibrated in accordance with MIL-STD-810, Method 514.2, Procedure VIII, except the test level and frequency range shall be as specified in 3.9.1. Vibration shall be applied along each of three mutually perpendicular axes of the body assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the body assembly along three mutually perpendicular axes while the body assembly is maintained in its

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normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing 10516576). The body assembly shall be clamped on the vibrating equipment by means of an appropriate adapter that has the necessary mounting features (key ways and locating surfaces). The body assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period indicated in 3.9.1. Immediately following the vibration cycle the body assembly shall be subjected to the tests in 4.6.4.5 and 4.6.4.6. Body assemblies that have passed the above tests shall then be subjected to the tests specified in 4.6.3 through 4.6.4.1 inclusive and 4.6.5 through 4.6.10 inclusive, to determine overall conformance with 3.9.1.

NOTE 1: Caution shall be exercised during transfer and handling of the body assembly, between tests, so as not to disturb the boresight knob scale settings of (4).

NOTE 2: The above tests for 4.6.3 shall be conducted at a temperature of $73^{o} \pm 18^{o}F$.

4.6.4.3 Vibration "B". This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope, this vibration test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the body assembly shall be subjected to the test procedure specified in 4.6.4.1 to determine compliance with 3.8.1. Subsequent to the test in 4.6.4.1 the body assembly shall be vibrated in accordance with MIL-STD-810, Method 514.2, Procedure VIII, except the test level and frequency range shall be as specified in 3.9.2. Vibration shall be applied along each of three mutually perpendicular axes of the body assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the Tolerance of Test Conditions specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the body assembly along three mutually perpendicular axes while the body assembly is maintained in its normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing 10516576). The body assembly shall be clamped on the vibrating equipment by means of an appropriate adapter that has the necessary mounting features (keyways and locating surfaces). The body assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period indicated in 3.9.2. Immediately following the vibration cycle the body assembly shall be subjected to the tests in 4.6.4.5 and 4.6.4.6. Body assemblies that have passed the above tests shall then be subjected to the test specified in 4.6.3 through 4.6.4.1 inclusive and 4.6.5 through 4.6.10 inclusive, to determine overall conformance with 3.9.2. The caution notes 1 and 2 of 4.6.4.2 again apply.

4.6.4.4 Shock. This test shall be performed on body assemblies procured as a repair part item. When the body assembly is provided with a complete periscope, this shock test will not be required when the complete periscope is subjected to shock. Prior to performing this test, the body assembly shall be subjected to the test procedure specified in 4.6.4.1 to determine compliance with 3.8.1. The shock equipment and test procedure shall be in accordance

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with MIL-STD-810, Method 516.2, Procedure 1, except conditions such as "G" value, time duration and wave form shock pluse shall be as specified in 3.9.3. The body assembly shall be clamped on the shock test equipment by means of an appropriate adapter that has the necessary mounting features (keyways and locating surfaces). The body assembly shall then be shocked in accordance with the test procedure specified above, except the application of shock along the vertical axis shall be in one direction only (normal operating position of the body assembly). Immediately following the shock test the body assembly shall be subjected to the tests in 4.6.4.5 and 4.6.4.6. Body assemblies that have passed the above tests shall then be subjected to the tests specified in 4.6.3 through 4.6.4.1 inclusive and 4.6.5 through 4.6.10 inclusive, to determine overall conformance with 3.9.3. The caution notes 1 and 2 of 4.6.4.2 again apply.

4.6.4.5 Post-vibration/post-shock examination. Subsequent to the vibration/shock test the body assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.9, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. With the exception of the slip scales, a tactile examination shall be made for functioning of moving parts for undue irregularities of movement, friction or looseness. An audible examination shall also be made of the body assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a body assembly defective. Body assemblies which have passed the above examinations shall be subjected to the tests speicfied in 4.6.4.6.

4.6.4.6 Post vibration/post-shock collimation change. Subsequent to the test in 4.6.4.5 the body assembly shall again be subjected to the test procedure specified in 4.6.4.1. Any difference from the recorded boresight knob scale settings established in 4.6.4.1 shall not exceed the tolerance specified in 3.10, as measured on the boresight knob scales.

NOTE: Do not slip knob scales.

4.6.5 Orientation. The following tests shall be performed with the body assembly mounted on special testing equipment conforming to 4.5.7.1.2. The target projector collimator shall be adjusted for 1200 meters and unless otherwise specified, the eyepiece diopter knob shall be set for best focus of the body assembly reticle and the target image.

4.6.5.1 Parallax. Position the body assembly reticle boresight mark as in 4.6.4.1. Parallax requirements shall be met when the eccentric is adjusted as specified in 3.7. While viewing through the body assembly eyepiece parallax will be recognized by any apparent displacement of the intersecting point of the reticle boresight mark with respect to the intersecting point of the vertical target line and the horizontal target line located 20 mils left of the geometric axis of the targe reticle when the observer's head is moved from side to side or up and down. Displacement between the body assembly reticle and target image shall not exceed the tolerance specified in 3.7. If adjustment is necessary see 4.6.5.1.1. Upon satisfactory completion of this test (parallax), perform the tests specified in 4.6.5.9 and 4.6.3.4.

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4.6.5.1.1 Parallax adjustment. If any adjustment is necessary, a standard type screw driver that properly fits the screw slot of the locking screw on Housing 11727432 and Eccentric 8205635 shall be utilized for performing this adjustment. Back off the locking screw a minimum of two revolutions from its locked position. Then while performing the test in 4.6.5.1, rotate the eccentric clockwise or counterclockwise until the parallax requirement specified therein has been met. When the adjustment is complete, apply the applicable torque value specified in 3.5.4 to the locking screw utilizing the torque wrench specified in 4.6.3.4.

4.6.5.2 Reticle movement, elevation. Position the body assembly reticle boresight mark as in 4.6.4.1 (do not slip knob scales). While observing the boresight mark (intersecting point), rotate the elevation knob in the clockwise direction. The boresight mark shall move as specified in 3.8.2.1 as observed and measured on the target. Re-establish coincidence of the reticle boresight mark and the projected target image crosslines by means of the boresight knobs. While observing the boresight mark shall move as specified in 3.8.2.1 as observed and measured on target.

4.6.5.2.1 Reticle movement, deflection. Perform this test utilizing the procedure in 4.6.5.2, except the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight mark (intersecting point) movements shall meet the requirements specified in 3.8.2.2 as observed and measured on the target.

4.6.5.3 Knob rotation. Perform this test in conjunction with the tests in 4.6.5.2 and 4.6.5.2.1. During rotation of each boresight knob (elevation and deflection), the boresight mark (intersecting point) shall be observed and show no evidence of irregularities (jump, creep or hang up) in direction of travel until limited by engagement of each knob stop (clockwise and counterclockwise) to be in compliance with 3.8.3. While the boresight knobs are at each stop position, back off the knobs for at least 1/4 rotation and then again return the knobs to the stop. Perform this a minimum of 3 times. During these rotations, the knobs shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching the stop. Contact with the stop shall be solid, such that a distinct click can be heard to verify that reticle travel is being limited only by means of the knob stops and not by means of contact with adjacent components to determine overall compliance with 3.8.3.

4.6.5.4 Line of sight variation. Perform this test in a manner similar to the test in 4.6.5.2.1. During rotation of the deflection knob, any variation in elevation in the line of sight, through the boresight mark (intersecting point), shall not exceed the values specified in 3.8.4.

4.6.5.5 Reticle shift. With backlash removed for direction of travel, position the body assembly reticle boresight mark as in 4.6.4.1. While observing the intersecting point of the body assembly reticle boresight mark, disengage the elevation boresight knob from the lock position and proceed to rotate the knob (insure backlash is not introduced) for at least 1/4 turn,

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then reverse rotation of the knob (backlash included) for at least 1/4 turn. During the disengagement or reversal of the boresight knob, the line of sight through the body assembly reticle boresight mark shall not shift (jump) in a direction perpendicular to the intended motion more than the tolerance specified in 3.8.4.1. In addition to checking shift at this setting, repeat the above at the midpoint and maximum point of travel in each direction for a total of 5 checks overall. These 5 checks shall also be conducted with the deflection knob in compliance with 3.8.4.1.

4.6.5.6 Boresight backlash. While viewing through the body assembly eyepiece, disengage and rotate the boresight knobs (insure backlash is removed for direction of travel) until a line of sight, through the intersection of the body assembly reticle boresight mark is positioned as in 4.6.4.1. With the boresight knobs in the lock position, slip the scales to Index 4 and record the position of the line of sight. Disengage the boresight knobs and continue rotation approximately 1/2 of a revolution then return and engage the knobs to their original lock position setting of Index 4. View through the eyepiece and record the position of the line of sight through the reticle boresight mark. Any difference between the recorded positions of the line of sight for each boresight knob is the amount of backlash. Backlash in each boresight mechanism shall not exceed the tolerance specified in 3.8.5, as measured on the target.

NOTE: Should overtravel occur during rotation of the boresight knobs the test is nullified and shall be repeated.

4.6.5.7 Image tilt. This test shall be performed with the aid of the observation telescope positioned at the body assembly eyepiece. View through the observation telescope and establish one extremity of the telescope horizontal reticle line coincident to the horizontal target image line. The image of the horizontal target line shall be parallel to the telescope vertical line within the tolerance specified in 3.11, as measured on the observation telescope.

4.6.5.8 Parallelism of image and reticle. Perform this test while observing through the body assembly eyepiece with the unaided eye, position one extremity of the body assembly vertical reticle line coincident to the vertical target image line. The body assembly vertical reticle line shall be parallel to the vertical target image line within the tolerance specified in 3.12, as measured on the target.

4.6.5.9 Objective focusing range. This test shall be performed subsequent to the test in 4.6.5.1. A dial indicator, graduated in .001 of an inch, mounted on a appropriate stand, a supporting medium capable of receiving the dial indicator as well as the body assembly to be tested and the screw driver specified in 4.6.5.1.1 shall be utilized for this test. With the body assembly firmly secured on the supporting medium, position the dial indicator so that its contact point touches the face of the moveable objective tube of the body assembly and set the dial to read zero with the indicating hand. Back off the locking screw of Housing 11727432 a minimum of two full revolutions from its locked position. Starting from the zero reading of the dial

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indicator, rotate the eccentric until the objective assembly has traveled the minimum distance specified in 3.13, as measured on the dial indicator. Return the indicating hand of the dial indicator to zero by means of the eccentric and repeat this travel in the opposite direction. there shall be sufficient travel in each direction to meet the minimum requirements specified in 3.13, as measured on the dial indicator. Upon completion of this test, repeat the parallax adjustment procedure specified in 4.6.5.1.1. The body assembly shall then be subjected to the test in 4.6.3.4.

4.6.5.10 Dioptral adjustment.

4.6.5.10.1 Zero setting. A calibrated dioptometer with a magnification of at least 3 power (3X) shall be utilized for measuring the accuracy of the zero diopter setting of the diopter scale. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Position the dioptometer at the eyepiece of the body assembly. While viewing through the dioptometer, obtain best focus of the body assembly reticle by adjusting the body assembly diopter knob. The diopter scale shall indicate zero within the tolerance specified in 3.14.1.

4.6.5.10.2 Excursion range. Perform this test in conjunction with the test in 4.6.5.10.1. From the best focus of the body assembly reticle established in 4.6.5.10.1, rotate the diopter adjustment knob to its clockwise stop, note the diopter setting on the scale, then rotate the diopter knob to its counterclockwise stop, note diopter setting on the scale. The diopter movement shall meet the requirements specified in 3.14.2, as measured on the diopter scale of the body assembly.

4.6.5.11 Diopter backlash. Perform this test utilizing the dioptometer specified in 4.6.5.10.1. The dioptometer shall be zeroed and adjusted as specified therein. Starting from the stop position rotate the body assembly diopter adjustment knob clockwise to the "0" position without reversing direction of rotation. Position the dioptometer at the eyepiece and obtain best focus of the body assembly reticle by adjustment of the dioptometer diopter scale. Record the dioptometer scale reading. Rotate the body assembly diopter adjustment knob clockwise to its stop position, then return it counterclockwise to the "0" position without overtravel. Again obtain best focus of the body assembly reticle by adjustment of the dioptometer diopter scale. The difference between the first and second dioptometer diopter scale readings shall not exeed the value specified in 3.14.3.

NOTE: Should overtravel occur during rotation of the diopter adjustment knob to the "0" position, the test is nullified and shall be repeated.

4.6.6 Resolution. The test equipment specified in 4.6.5 is not required for this test. Perform this test utilizing the dioptometer specified in 4.6.5.10.1. The dioptometer shall be zeroed and adjusted as specified therein. A resolving power chart on a wall target representing the angular subtense for six seconds of arc and a supporting medium capable of receiving

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the dioptometer (as well as the body assembly to be tested) shall be a part of the test equipment. Construction of the resolving power chart shall be as specified in the "Resolution Test" of MIL-O-13830 and shall contain four line sets as shown in Figure I therein. Place the body assembly on the supporting medium and position the dioptometer at the eyepiece. While viewing through the dioptometer, obtain best focus of the numeral in the center of the resolving power chart by adjusting the body assembly diopter knob, then obtain best focus of each of the four meridians by adjustment of the dioptometer objective scale. Any change in the diopter readings by adjustment of the dioptometer objective scale to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the dioptometer objective scale. All four meridians shall have the detectable line structure and proper line count to conform with 3.15 and the difference between the dioptometer scale readings for any two meridians shall not exceed the value specified therein.

4.6.7 Reticle illumination. This test shall be performed in a dark area. The voltage required to illuminate the reticle lamp shall be supplied from a power source through a control device that shall be capable of supplying and maintaining the required voltage in 3.16. With the reticle lamp energized, the illuminated reticle markings shall be clearly defined when observed through the body assembly eyepiece to determine conformance with 3.16. Should the brightness of a reticle defect be greater than the reticle lines, the defect shall be cause for rejection.

4.6.8 Sealing. This portion of the test shall be performed utilizing a pressure gage capable of reading between 0.25 and 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall be adhered to. Check the internal pressure of the body assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the body assembly by means of the dew point tester. Interior atmosphere of the body assembly shall conform to the requirements of 3.17. Upon satisfactory completion of this portion of the test, internal pressurizing of the body assembly shall be performed with special testing equipment conforming to 4.5.7.1.1. Pressurize the body assembly with dry nitrogen gas, having a dew point at least as low as -25°F, to 5 $\pm$ 0.10 psig in accordance with the procedure for internal testing on Drawing 8565556. Retain the applied pressure in the body assembly for the time duration specified in 3.17. Within the first 60 minutes of this time period, the boresight knobs and the diopter adjustment shall be rotated to their clockwise and counterclockwise stops, and with the locking screw on Housing 11727432 backed off from its locked position, the Eccentric 8205635 shall be rotated a minimum of two full revolutions in each direction. At the end of this time period, any evidence of leakage as indicated on the pressure gage of the testing equipment shall not exceed the tolerance specified in 3.17. Upon satisfactory completion of the sealing test, the body assembly shall then be flushed with the dry nitrogen gas specified above. The interior atmosphere of the body assembly shall be periodically sampled through the dew point tester until the atmosphere shows a dew point at least as low as -25°F. When this atmosphere is attained within the body assembly, the body assembly shall then be sealed at the pressure specified in 3.17. Upon completion, readjust the eccentric to the position established in 4.6.5.1 and torque the locking screw as specified in 3.5.4.

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4.6.9 Cleanliness. The optical surfaces of the body assembly shall be visually examined with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the eyepiece, inspection for dirt shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot candles. The body assembly shall conform to the requirements of 3.18.

4.6.10 Interchangeability. The interchangeability test, of the body assembly to head assembly, shall be performed utilizing the maximum and minimum interchangeability gages conforming to Drawings 11732109 and 11732110. The body assembly shall conform to the requirements of 3.2.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with the Packaging Data Sheet 10516576. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Body Assembly 10516576 is a daylight modular unit of the Tank Gunner's Periscopes M32C and M32CE1, primarily used in the Fire Control System of the 165MM Gun, Combat Engineering Vehicle: M728.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Quantity required and delivery schedules.
c. Applicable packaging data sheet number (see 5.1).
d. Selection of applicable levels of preservation, packaging and packing.
e. Applicable stock number.
f. Requirements for First Article Submission, including data requirements for certificates of conformance for materials.

6.3 Inspection equipment design.

6.3.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with the technical data listed in the Technical Data Package (TDP).

6.3.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.6 of this specification and with the "Test Conditions" and "Test Facilities and apparatus" specified in the "General Requirements" of MIL-STD-810. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product

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tolerance limit. Thus, concept, construction, materials and dimensions shall be so selected and controlled as to insure that the test equipment will reliability indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabrication or procuring the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit as required to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAF-I, Dover, NJ 07801. When the contractor submits inspection equipment designs to the Government of approval, he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.)
c. The designs remaining to be submitted and the expected date of submittal.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading of U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical date originally prepared by these activities is now under the cognizance of ARRADCOM.

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Preparing Activity:

Army - AR

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